Simulator Reference Data and Sources

October 25, 2026 · HackIoT classroom AC simulator (`apps/web/`)

The simulator's energy and cost figures take each hour's cooling load, scale it by a published "about 10% less per 1°C warmer" factor, and price it with LADWP rates. This document lists where every number comes from and how it is used.

At a glance

Each number is one of three kinds: from the literature (citable), from LADWP's official rates, or a demo assumption we chose. All code paths are under `apps/web/`.

Number	Value	Code	Kind	Source
AC electricity per 1°C warmer room	10% less (× 0.9)	`lib/model-sim.ts` `ENERGY.savingPerC`	Literature	Hoyt, Arens & Zhang 2015
Heat per person (cooling load)	72 W sensible + 45 W latent = 117 W	`ENERGY.sensibleW`, `latentW`	Literature	ASHRAE Fundamentals Ch. 30 Table 1
Electricity price (by month and tier)	$0.24362–0.40968 /kWh	`lib/billing.ts` `tierRates()`	Official rates	LADWP R-1A 2026
Tier limits	700 / 2,100 kWh per two-month bill	`TIER_LIMITS_KWH`	Official rates + our doubling	LADWP rate ordinance, Zone 1 × 2
Lights and plug loads	10 W/m²	`ENERGY.internalWPerM2`	Our assumption	None
Envelope heat loss	1.8 W/m²·K	`lib/thermal.ts` `lossWPerM2K` (slider)	Existing repo assumption	None
AC efficiency (COP)	3	`ENERGY.cop`	Existing repo assumption	None
Carbon factor	0.4 kg CO_2e/kWh	`ENERGY.kgCo2PerKwh`	Existing repo assumption	None
Reference room temperature for the load	24°C	`FIXED_SETPOINT_C`	Our assumption	None
Baseline for comparison	AC fixed at 22°C	`BASELINE_AC_C`	Chosen by the team	Close to Hoyt's 22.2°C baseline
AC when the room is empty	26°C	`SETBACK_C`	Chosen by the team	None
How far people lift the room above the AC setpoint	−2 + 8·(N−1)/299 °C	`heatLiftC()`	Derived from the model's prior	Team's `acmodel.py`
Heat per person (for the room lift)	100 W	`thermal.ts` `wattsPerPerson` (slider)	Existing repo assumption	None
Outdoor temperature and humidity	27°C, 50%	`lib/sim-rooms.ts` `scenarioFor()`	Demo value	None

Number	Value	Code	Kind	Source
Student preference spread and voting threshold	SD 2°C, vote when ±1°C off, half of them vote	`simulatedVotes()`	Our assumption	None
Room sizes, capacities, timetables	6 rooms	`lib/sim-rooms.ts`	Demo data	None

Temperature: about 10% less AC electricity per 1°C

We use Hoyt et al. (2015): raising the cooling setpoint from 22.2°C to 25°C (2.8°C) cut cooling energy by 29% on average and total HVAC energy by 27%. Per degree, we round to 10% and compound it:

$$E(T_{room}) = \frac{Q_{24}}{COP} \times 0.9^{\,(T_{room}-24)} \times h$$

- `Q_24` is the heat the AC removes with the room at 24°C (kW), built up in the next two sections. `h` is the number of hours (1).
- Savings = baseline electricity − model electricity. Both use this formula and differ only in room temperature.
- Example: an AC at 25°C is 3°C warmer than the 22°C baseline, $0.9^3 = 0.729$, so 27% less. That is the "uses 27% less" shown for the hour.

Conversion: 29% over 2.8°C is about 10.4% per °C (linear), or (1 − 0.29)^(1/2.8) ≈ 11.5% per °C (compounded). We use 10%, below both, so the savings estimate leans conservative.

Source	Finding	How we use it
Hoyt, Arens & Zhang, *Building and Environment* 88 (2015)	22.2→25°C: cooling energy −29%, HVAC −27% (DOE medium office models, seven US climate zones)	Converted to 10% per °C; the main basis for `savingPerC`
Ghahramani et al., Winter Simulation Conference (2015), USC	Cites Hoyt's 29% / 27%. Own results: widening the deadband from 3 K to 6 K saves 16.2%; optimal annual setpoints save 6.63% on average (Los Angeles climate included)	The source we opened to check the 29%; citable as USC research in the presentation
de Chalendar et al., *Energy and Buildings* 278 (2023)	Field data, six commercial buildings: +2°F (1.1°C) cut cooling loads 13–28% in offices and 3–4% in labs; **effect on electric loads under 2%**	Supports the range only; not in the formula
Cardiff University (ORCA repository)	Simulation: about −8% cooling energy per +1°C	Only in a code comment. The page did not open (403) and the number came from a search summary, so do not cite it

Note: these studies are of office buildings and measure the thermostat (room) setpoint. We apply them to classrooms, and in the simulator the AC setpoint and the room temperature are separate, so the formula uses the **room** temperature (see the heat lift below).

People: 117 W each, so more people means more kWh per degree

Head count enters twice: in the cooling load (a published number) and in how far the room sits above the AC setpoint (our assumption).

1. Cooling load: 72 W sensible + 45 W latent per person

ASHRAE Fundamentals Ch. 30 Table 1, "seated, very light work": 245 Btu/h sensible and 155 Btu/h latent, which is 72 W + 45 W = 117 W. Students in class fit this activity level. The AC also removes the latent heat (moisture from breathing), so both count.

$$Q_{people} = N \times (72 + 45)\ \mathrm{W}$$

Example: a 260-person lecture hall adds 30.4 kW; a 36-person classroom adds 4.2 kW. The same 1°C saves several times more kWh in the lecture hall, which is what "more people means more heat to remove" refers to.

2. How far the room sits above the AC setpoint (heat lift)

$$T_{room} = T_{AC} + \left(-2 + \frac{8\,(N-1)}{299} \times \frac{W}{100}\right)$$

- Source: the prior curve in the team's `acmodel.py` (26°C for 1 person, 18°C for 300). We assumed the room sits at 24°C whenever the AC follows that prior, which gives this formula.
- `W` is the "Heat from one person" slider (default 100 W). It only drives the room lift and is separate from the 117 W in the load.
- This is our own assumption, not from the literature. It puts the room 4.9°C above the setpoint with 260 people, and 2°C below with 1 person.

Floor area: lights, plugs and the envelope

Floor area drives two loads. Both coefficients are assumptions, not from the literature. The full hourly load:

$$Q_{24} = \underbrace{117N}_{\text{people}} + \underbrace{10A}_{\text{lights, plugs}} + \underbrace{1.8A\ \max(0,\ T_{out} - 24)}_{\text{envelope}} \quad [\mathrm{W}]$$

Item	Coefficient	Code	Notes
Lights and plug loads	10 W/m²	`ENERGY.internalWPerM2`	Rough estimate for classroom lighting, projectors and laptops
Envelope heat loss	1.8 W/m²·K	`lossWPerM2K` ("Envelope loss" slider)	Heat gain per m² of floor per degree of indoor/outdoor difference, simplified to floor area
Outdoor temperature	27°C	`scenarioFor()`	Fixed demo value, adjustable with a slider

Example: Leavey Reading Room, 640 m², empty: 6.4 kW lights and plugs + 3.5 kW envelope = 9.9 kW, divided by COP 3 ≈ 3.3 kWh per hour (room at 24°C). An empty room still uses power, which is why setting it back during empty hours accounts for a large share of the savings.

COP 3 (1 kWh of electricity removes 3 kWh of heat) and 0.4 kg CO_2e/kWh come from the repo's existing `lib/energy-impact.ts`, which labels them illustrative inputs, not measured equipment ratings.

Cost: LADWP R-1A, tiered and billed every two months

Each hour's kWh is added to the current bill in time order and priced at that month's rate for the tier it lands in. Each bill (Nov–Dec, Jan–Feb, …) starts again at Tier 1. The baseline and the model each get their own bill; dollars saved = the difference.

2026 total consumption charges including adjustment factors ($/kWh), from LADWP's site:

Months	Tier 1	Tier 2	Tier 3
Jan–Mar	0.24771	0.30630	0.30630
Apr–May	0.24362	0.30221	0.30221
Jun	0.24362	0.30221	0.38922
Jul–Sep	0.26408	0.32267	0.40968
Oct–Dec	0.27292	0.33151	0.33151

Tiers: LADWP's rate ordinance gives monthly allocations of 350 / 1,050 kWh for Zone 1 and 500 / 1,500 kWh for Zone 2. We doubled Zone 1 for a two-month bill: 700 / 2,100 kWh.

To fix: USC is in Zone 2. The ordinance's zip code table lists 90007 (USC's campus) under Zone 2, so the Zone 1 in the code comment is wrong. With Zone 2, the two-month tiers would be 1,000 / 3,000 kWh. Tiers 1 and 2 get wider and the dollar savings come out slightly lower. The "double it for two months" rule also came from a third-party site; we have not found it in LADWP's own documents yet.

Left out: the monthly Power Access Charge (both bills sit in the top tier, so it cancels) and the 10% city utility tax. Months in 2027 reuse 2026 rates. A real campus is a commercial account, billed monthly with demand charges; R-1A is a residential rate, used here to match the two-month billing period.

Scale-up "per room per day": a copy of the model runs the next 8 weeks (1,344 hours), priced as above, then divided by 56 days (`lib/sim-run.ts` `dailySavingsUsd()`). Total = per day × rooms × days, a straight multiplication with no seasons.

The model and the simulated votes

The AC setpoint comes from the team's `apps/alg/acmodel.py` (the frontend's `lib/ac-model.ts` is a port checked hour by hour against it). Its parameters were chosen by the team, not taken from the literature.

Parameter	Value	What it does
Prior curve	1 person 26°C → 300 people 18°C (straight line)	The setting before anything is learned
AC step	0.5°C, range 16–30°C	Rounds the output

Parameter	Value	What it does
Hysteresis	0.3°C	Changes under 0.55°C are ignored
Vote correction	atanh(0.95 × vote margin) × 2.5, at most ±5°C	Majority rule: more hot votes means cooler
Curve decay	× 0.98 per hour	Old data fades
Group offset decay	× 0.9 per week, limit ±7°C	Each group's own preference
Group detection	≥ 15 people; same weekday, same start hour, head count within max(5, 25%)	Recognizes the same group the next week

Simulated votes (`simulatedVotes()`, simulator only, our assumption): each student's preferred room temperature is normally distributed around the group's "Group prefers" value with SD 2°C. Students whose preference is more than 1°C below the room want to vote hot, more than 1°C above want to vote cold, and half of them actually vote.

$$hot = \text{round}\left(N \times 0.5 \times \Phi\left(\frac{T_{room} - 1 - T_{want}}{2}\right)\right), \quad cold = \text{round}\left(N \times 0.5 \times \left[1 - \Phi\left(\frac{T_{room} + 1 - T_{want}}{2}\right)\right]\right)$$

`Φ` is the standard normal CDF, computed with the Abramowitz & Stegun 7.1.26 approximation. Each room's preferred temperatures (22–26°C) are demo values, not survey data.

Assumptions to explain in a presentation

The energy and cost figures are estimates based on the literature, not measurements. These are the points people will ask about.

- **10%/°C comes from office-building simulations.** de Chalendar (2023) measured lower cooling loads but under 2% change in whole-building electricity. We estimate AC electricity only, not the whole building.
- **The baseline is an AC fixed at 22°C around the clock.** In empty hours the model sets back to 26°C, so much of the savings comes from raising the temperature when nobody is there, not from matching preferences.
- **The AC setpoint is not the room temperature.** Room = setpoint + heat lift, derived from the model's prior with no published source.
- **10 W/m², 1.8 W/m²·K, COP 3, 0.4 kg CO_2e/kWh and the 27°C outdoor temperature** are illustrative values.
- **The price uses a residential rate** and currently uses Zone 1 tiers by mistake; USC should be Zone 2.
- **Rooms, timetables and each group's preferred temperature are demo data**, not real USC classrooms or surveys.
- **The home, Impact and Dashboard pages use a different, older formula** (`lib/thermal.ts`: kWh = |envelope heat – people heat| × hours, at $0.25/kWh), so their numbers will not match the simulator's.

Sources

Source	Used for	Checked
Hoyt, Arens & Zhang, "Extending air temperature setpoints: Simulated energy savings and design considerations for new and retrofit buildings", *Building and Environment* 88 (2015)	10%/°C	Numbers confirmed through the next two sources; the paper's own page returned no content
Ghahramani, Dutta, Yang, Ozcelik & Becerik-Gerber, "Quantifying the influence of temperature setpoints, building and system features on energy consumption", WSC 2015 (USC)	Citation of 29% / 27%	Full text opened
CBE Setpoint Energy Savings Calculator, UC Berkeley	Confirms the Hoyt citation	Opened
de Chalendar, McMahon, Fuentes Valenzuela, Glynn & Benson, "Unlocking demand response in commercial buildings", *Energy and Buildings* 278 (2023)	Supports 10%/°C	Abstract opened
ASHRAE Fundamentals Ch. 30 Table 1, as cited by Autodesk Revit help	72 W + 45 W	Opened; the ASHRAE handbook itself was not checked
LADWP Residential Rates	2026 monthly prices	Opened
LADWP Electric Rates (rate ordinance PDF)	Zone 1 / 2 tiers and zip code zones	Opened; 90007 is in Zone 2
Cardiff University ORCA, record 154774	8%/°C (code comment only)	Not opened (403); number from a search summary, do not cite
Team's `apps/alg/acmodel.py` (hapiwangy)	Prior curve and model parameters	Repo main